

RAPHAEL MARIO ROCHA

Software Engineer
AI Application



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<https://raphael-rocha.xyz> • Visa Status: **F-6**

French: Native

English: Fluent

Korean: Conversational (TOPIK 3~4)

Education

University Marie et Louis Pasteur (UMLP)

Bachelor's Degree - Computer Science

Belfort, France

Graduation: 2021

Summary

French software engineer living in South Korea for over 3 years, with 2 years of professional experience. Worked for a year at a Seoul-based B2B SaaS startup in 2025.

Area of expertise:

- **AI Application:** LangChain, RAG, Qdrant, LangSmith, Vertex AI (Gemini), Claude
- **Backend:** Python / Django (DRF), Flask, Fast API, Java / Spring
- **DevOps:** Docker, Git, Github, Gerrit, Ansible
- **Cloud & Infra:** AWS, Cloudflare
- **Frontend:** Vue.js, Vuex, Tailwind
- **Database:** PostgreSQL

My interest in mathematics and computing naturally led me to software engineering, where I enjoy solving complex problems through logic, algorithms, and well-structured systems. I follow strong engineering principles and make sure requirements are clearly understood so that every decision contributes to clean, efficient, and maintainable solutions.

Beyond making software work, I focus on building systems that reduce technical debt, improve reliability, and scale with the needs of the product. In recent years, I also integrated AI-assisted development tools into my workflow to improve efficiency while maintaining strong technical judgment and remaining mindful of long-term maintainability and code quality.

My goal is to bring clarity, structure, and long-term thinking to a team so I can contribute to a healthier codebase, faster development cycles, and the overall success of the company.

Working Experience

PhyxUp Health

Software Engineer

SaaS in Medical/Therapy Industry in US (<10 employees)

help therapists providing a Remote Therapy Monitoring service to patients and manage the billing system.

Seoul, South Korea

02/2025 – 12/2025

- Developed and maintained features across both the **frontend** and **backend**, contributing to the core business logic, billing system and application performance.
- Designed and implemented API endpoints, worked on the **authentication** system of the app, logging and monitoring, enhanced code quality of legacy modules through structured refactoring, **SOLID** principles, and introduced security and performance improvements.
- Managed role and permission systems, implemented **WebSocket** communication, and integrated logging and compliance mechanisms aligned with HIPAA and FHIR standards.
- Worked with **PostgreSQL** on Amazon **RDS**.
- Managed **AWS** infrastructure, set up a new staging environment, and defined the deployment pipeline and Git workflow, with **Cloudflare** managing DNS.
- Evaluate key performance indicators, user interactions, and system metrics. Provide data-driven feedback to improve functionality, usability, and reliability.
- Improved **CI/CD** pipelines for both frontend and backend, managed mobile app packaging, versioning, and release for iOS and Android using GitHub Actions and Codemagic.
- Managed all services through **Docker** containers to ensure consistency across environments.
- Create structured documentation covering the application architecture, infrastructure, and key workflows, ensuring new and existing team members can quickly understand the system, onboard smoothly, and access step-by-step guides, best practices, and useful resources.
- Collaborated with an **Agile** team, regularly contributing ideas and feedback to improve the product and team workflows.

Backend: Django (DRF), Sockets, Redis

CI/CD: Docker, Git, Github, Github Actions, Codemagic

Cloud: AWS Services (EC2, RDS, S3...), Cloudflare (DNS, Pages/Workers)

Frontend : VueJs, Vuex, Tailwind, Vite/npm

Database: PostgreSQL

Atol Advice & Development (Atol CD)

Software Engineer

IT Consulting Company (~300 employees)

Dijon, France

06/2023 - 07/2024

- Contributed to multiple projects across different technology stacks (**monolithic** and **microservice** architectures), handling feature development, optimization, and bug fixes across both **frontend** and **backend**.
- Frontend: Built and maintained interfaces using **Vue.js** and TypeScript.
- Backend: Developed business logic and RESTful web services using Java **Spring**, JAX-RS, and JOOQ; implemented security improvements and on-premise CAPTCHA systems; worked with OpenLayers and **Geoserver** for geospatial features.
- Database: Used PostgreSQL with PostGIS for spatial data and Flyway for schema migrations.
- **CI/CD**:
 - Created and maintained Jenkins pipelines for build, release, and code review processes.
 - Wrote **Docker** based packaging scripts and handled RPM generation for legacy systems.
 - Integrated with Git and Gerrit for version control and code review workflows.
- Managed and upgraded demo server environments, upgraded Java versions and related dependencies, defined new packaging and deployment processes.
- Automated **Full server** setup using Ansible (Apache, SSH, etc.).
- Improved team workflow, project documentation, and deployment/versioning strategies.

Backend: Spring, Jax-RS, Apache, Tomcat, Maven
CI/CD: Docker, Git, Gerrit, Jenkins, Nexus Repository
Frontend: VueJs, Vuex, Tailwind, Vite/npm
Database: PostgreSQL, PostGIS

Other Experience in Korea

Manpower Korea

Seoul South Korea

Part time: Worked as a Headhunter for IT positions (*during a Working Holiday year*)

12/2022 - 05/2023

- Conducted proactive headhunting, candidate outreach, and screening interviews to evaluate technical fit and ensure alignment with client requirements, on Backend engineers, DevOps, Cloud, Network, NLP...

Internships

Departmental Council of Haute-Saone

Vesoul, France

Intern Web Developer (7 months)

10/2019 - 12/2021

03/2021 - 06/2021

Stack: Javascript, JQuery, PostgreSQL, PHP/PHPMyAdmin, html/css

Personal Projects

AI-Assisted Resume/JD Comparator | [Github](#) | [Live demo](#)

- Built a production-ready AI-assisted Resume/JD platform with two features: a matching feature that finds the best candidates for a given Job Description from the resume pool, and a comparison feature that scores fit between a single resume and JD.
- Implemented a "find best candidates" feature using a two-stage RAG pipeline (fastembed/**ONNX-runtime** embeddings): **Qdrant** vector search pre-filters the full resume pool to a shortlist, then an LLM scores each, limiting expensive AI calls to relevant candidates only.
- Paired Qdrant (vector embeddings) with SQLite (raw document store) to support both semantic retrieval and structured data access.
- Integrated LLM models via **LangChain** with Ollama (hosted locally on docker) or external Google Vertex AI (Gemini) providers through a configurable provider abstraction layer.
- Added observability for LLM/API behavior with structured logging and **LangSmith** tracing to improve debugging and prompt iteration.

Backend: Flask

AI: LangChain, RAG, Qdrant, LangSmith, Gemini (Vertex AI)

CI/CD: Docker, Git, Github

Infra : AWS/Cloudflare

Frontend: VueJs, Vite/npm

Core competencies

Technologies

- **AI Application**
 - LangChain, Observability with LangSmith
 - RAG system, Qdrant
 - fastembed/ONNX-runtime embeddings
 - Gemini (Vertex AI), Claude
- **Backend**
 - Python / Django (DRF), Flask, Fast API, Java / Spring, Hibernate or JOOQ
 - REST APIs, WebSockets, Redis, Sentry...
 - Web server: Apache, Tomcat, Nginx
- **DevOps**
 - CI/CD : Git, GitHub, Ansible, Gerrit, Jenkins, Codemagic
 - Artifact Management: Nexus Repository
- **Cloud / Infra**
 - AWS: EC2, S3, ECR, RDS, Route 53, Load Balancer, Certificate Manager, Registrar...
 - Cloudflare: DNS, Pages, Workers
- **Frontend**
 - Vue.js (Vuex), Three.js
 - Tailwind
 - Npm
- **Database**
 - PostgreSQL
- **Geospatial**
 - GeoServer, PostGIS, OpenLayers
- **Tools**
 - OS: Linux (Ubuntu)
 - Experience with AI Assistant tools, such as Cursor and Claude Code
 - DBeaver, Sublime Merge, Postman...

Soft Skills

- Adaptability
- Communication focused
- Always Learning
- Teamwork and listening